Sotto

Confidential Payout Infrastructure on Canton

A Protocol for Private, Auditable, Atomically-Settled Disbursement

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> *"Sotto" is a working name — a placeholder you can replace with one find-and-replace. It comes from sotto voce, "in an undertone": value moved quietly, in confidence, where only the right parties can hear it.*
>
> *Privacy should be a property of the rail, not a feature of the app.*

1. Abstract

Sotto is a confidential payout protocol on the Canton Network. It lets any organization — a company running payroll, a marketplace paying its sellers, a fund distributing to its portfolio, an aid agency disbursing to beneficiaries — pay many recipients at once, where each payment is visible only to its recipient, the complete batch is provable to a permissioned auditor or funder, and the whole batch settles atomically in a regulated dollar stablecoin. No competitor, counterparty, or member of the public can see whom an organization pays, how much, or how often.

Existing on-chain payout rails force a false choice. Public blockchains move money in seconds at near-zero cost, but they broadcast the entire payout ledger — recipients, amounts, frequency, the payer's roster and cost base — to anyone who looks. Traditional rails keep that data private, but they settle in days, cost multiples more, cannot clear a batch atomically, and cannot cryptographically prove to a regulator that funds reached their intended recipients. Confidentiality has been treated as a compliance feature bolted onto transparent systems, when it should be a property of the settlement layer itself.

Sotto makes it one. The protocol is built on Canton because Canton is the one production-scale network where transaction privacy is a property of the ledger rather than an overlay: only the parties to a transaction can see it, the network orders transactions without seeing their contents, settlement is atomic across all parties, and a regulator or auditor can be granted a real-time, read-only view of exactly the activity they supervise. Sotto turns these primitives into a single interface any application can call to run a confidential, auditable, atomically-settled disbursement. This litepaper describes the architecture, the confidentiality and compliance model, the developer interface, and the staged roadmap by which the protocol will ship.

2. The Problem

2.1 The Transparency-or-Nothing Gap

A business that pays people at scale produces, as a byproduct, one of the most sensitive datasets it owns: who it pays, how much, how often, and for what. Payroll reveals individual compensation. Vendor payouts reveal supplier terms and margins. Marketplace disbursements reveal seller earnings and the platform's take rate. Grant and aid payments reveal the identities of recipients who may be vulnerable. This data is commercially sensitive at best and physically dangerous to expose at worst.

Every general-purpose on-chain payout rail in production today is built on a transparent ledger. The money moves quickly and cheaply, but the payout record is public and permanent. A company that runs its payroll on a transparent chain has published its org chart and its salary bands. A marketplace that pays sellers on one has handed competitors its unit economics. The settlement layer leaks exactly the information the organization most needs to protect, and no amount of front-end design changes what the chain itself records.

2.2 What Traditional Rails Get Right, and Wrong

Correspondent banking and conventional payout providers do keep this data private — it sits inside a regulated institution's database, not on a public ledger. But that privacy is purchased at a steep price. Cross-border settlement runs through a chain of intermediary banks, each taking a cut and adding days of delay. A batch of payments is not atomic: some clear, some fail, some are reversed, and reconciliation becomes a manual exercise in chasing exceptions. And the privacy is opaque rather than provable — an auditor or donor cannot independently verify that funds reached their intended recipients without trusting the operator's own records.

So an organization that needs to pay many people across borders is left choosing between two incomplete options: a transparent chain that is fast and cheap but exposes everything, or a traditional rail that is private but slow, expensive, unprovable, and unable to settle a batch as a single unit.

2.3 The Missing Layer

The missing layer is a settlement rail that is **confidential by default and auditable by permission** — where confidentiality is not a feature an application tries to add on top of a transparent ledger, but a property of the ledger itself, and where the same rail can selectively prove a disbursement to exactly the parties entitled to see it.

On a transparent chain, privacy can only be approximated by working against the grain of the system — mixers, off-chain settlement, obfuscation — each of which breaks composability, breaks compliance, or both. That is the wrong layer to solve the problem. Confidentiality belongs in the rail. Sotto is the payout rail built on the one network where it can be.

3. Why Now, Why Canton

A protocol of this kind could be built on any network whose ledger keeps transactions private from non-parties, settles atomically across counterparties, and admits a permissioned auditor. As of today, one production network has all three properties built in, and the surrounding pieces a payout rail needs have just shipped.

3.1 Privacy as a Property of the Ledger

Canton is a privacy-enabled public network whose smart-contract language, Daml, makes confidentiality a property of how data is modeled rather than something added afterward. Every contract on Canton declares who its signatories and observers are, and only those parties — never the wider network — can see its contents. The network's synchronizer orders transactions and guarantees their validity without ever seeing what they contain; transaction data is encrypted and delivered only to the relevant participants. This is the primitive Sotto is built on: a payment can be modeled so that it is visible to the payer and a single recipient and to no one else, while a separate, named auditor party is granted a read-only view of the whole.

3.2 A Confidential Dollar to Settle In

A payout rail needs a settlement asset, and as of late 2025 the right one exists on Canton. USDCx, the Canton-native form of Circle's USDC, is a fully-backed dollar stablecoin that settles with configurable, need-to-know privacy and is interoperable with USDC across other networks. It gives Sotto a regulated, dollar-denominated medium in which to clear payments — so that recipients receive a stable dollar value, and the payer's cash flow is never exposed on a public ledger. Network fees on Canton are denominated in dollars and paid in the network's utility token, and can be abstracted away from the people sending and receiving value.

3.3 Production-Readiness

Canton has crossed from experiment into production. Major financial institutions participate in the network, a global payments network has joined as a super-validator to bring privacy-preserving payments onto it, and — most directly relevant to Sotto — the first real-world private stablecoin payroll has already been executed on Canton, settling in minutes with salary details kept off the public ledger. That milestone proves the capability is real. It was also a single enterprise integration aimed at the top of the market. Sotto generalizes the same capability into a payout rail any organization can call, across every kind of disbursement, with auditability built in — and aims it at the segments that first integration did not serve.

3.4 The Build Window

The demand is not hypothetical. Stablecoin-settled cross-border payments are growing explosively, especially in emerging markets, where on-chain dollar volume has surged and surveyed recipients overwhelmingly prefer to be paid in stablecoins over local currency. The largest global payroll platforms have begun paying out in stablecoins. But that activity is running almost entirely on transparent rails, and the confidential layer beneath it has not been built. Every quarter it stays unbuilt, transparent payout rails harden into the default and the sensitive data they expose accumulates. Sotto exists because the network, the settlement asset, and the market have arrived at the same point at the same time, and the confidential payout layer between them is still open.

4. Protocol Architecture

4.1 Architectural Overview

Sotto is organized into three protocol layers, each addressing a distinct concern in the lifecycle of a payout. The separation is deliberate: the rules of *who can see* a payment, *how value moves*, and *who is allowed to disburse* evolve independently and do not cascade into one another. Above the protocol sit the surfaces that invoke it; below it sits the Canton Network.

Layer	Responsibility	Realized through
Disclosure	Determining who can see each payment and the batch as a whole; keeping payout data confidential by default; granting selective, read-only visibility to auditors and funders	Daml signatories and observers; Canton sub-transaction privacy; per-payment disclosure policies
Settlement	Moving value with finality; ensuring a batch clears as a single unit; tying release to conditions where required	Canton two-phase commit (atomic across all parties); USDCx settlement; atomic delivery-versus-payment
Authorization	Defining who may disburse, to whom, and under what limits; enforcing approvals before value moves	On-ledger mandate contracts; spending caps, allow-lists, and expiries; maker-checker approval thresholds

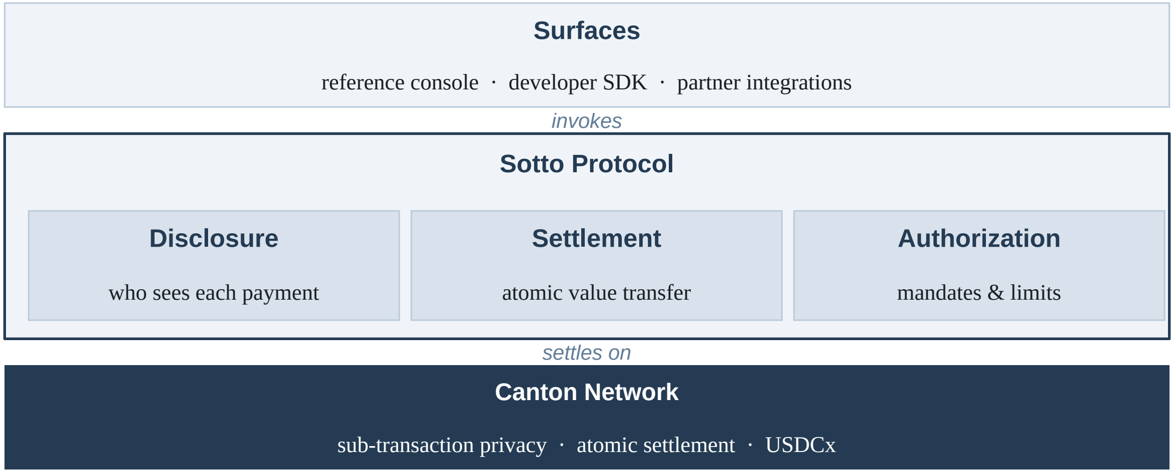

Figure 1. The Sotto protocol stack. Any surface invokes the same protocol; the protocol always keeps payments private to their parties, always settles atomically, and always enforces the payer's mandate, on Canton.

The guarantee that unifies them: a payout is constructed once, and the protocol enforces *simultaneously* that the right parties see it, that it settles atomically, and that it stays within the payer's authorized bounds — none of which any calling application can override.

4.2 The Disclosure Layer

The disclosure layer is the confidentiality core of the protocol, and it is where Sotto's central property lives. A payment in Sotto is a Daml contract whose stakeholders are exactly the payer and the single recipient it concerns. By construction, no other recipient in the same batch, no competitor, and no member of the public can see it — they cannot see the amount, cannot see the counterparty, and cannot see that the payment exists at all.

Visibility beyond the two principals is never incidental; it is granted explicitly through a disclosure policy attached to the batch. An auditor, a regulator, or — for a grant or aid program — a funder can be named as an observer with a real-time, read-only view of the entire batch, while remaining unable to alter or initiate anything. The result is a payout ledger that is private between counterparties and transparent to the parties entitled to oversight, with no middle ground that leaks to everyone.

4.3 The Settlement Layer

The settlement layer moves value with finality. Canton's two-phase commit means a transaction either commits across every stakeholder or is rolled back everywhere; there is no partial state. Sotto uses this to make a disbursement batch atomic: the payer can require that all payments in a batch settle together or none do, eliminating the half-paid, must-reconcile failure mode that plagues conventional batch payouts. Where a payout should be released only against a condition — proof of delivery, completion of a milestone, receipt of a counter-asset — settlement can be modeled as an atomic delivery-versus-payment, so value and obligation change hands in the same indivisible step. Settlement is denominated in USDCx, giving recipients a stable dollar value and finality measured in seconds.

4.4 The Authorization Layer

The authorization layer governs who is allowed to move money and under what constraints, and it enforces those constraints on the ledger rather than in any application. Before a payout can run, an organization establishes a **mandate**: an on-ledger contract that encodes a spending cap, a set of approved recipients, an expiry, and — above a configurable threshold — a maker-checker rule requiring a second authorizer to sign before value moves. Because the mandate is enforced by the contract itself, no calling application, and no individual operator, can exceed the cap, pay a recipient outside the allow-list, or push a large payment through without the required second approval. Authorization is a guarantee of the rail, not a policy the application is trusted to honor.

5. Confidentiality, Compliance, and Settlement Integrity

5.1 Disclosure Threat Model

Sotto's confidentiality model addresses three primary categories of exposure.

The first is **competitor and public surveillance** — anyone observing the ledger to infer an organization's roster, compensation, supplier terms, or cost structure. This is foreclosed by sub-transaction privacy: non-parties cannot see a payment's contents or its existence.

The second is **counterparty leakage** — one recipient learning what another recipient was paid. This is foreclosed by per-payment stakeholder scoping: each recipient is a party only to their own payment and has no visibility into any other.

The third is **operator overreach** — the protocol operator or a payer's own staff seeing or doing more than they should. This is constrained by the authorization layer (operators cannot move value outside a mandate) and by the disclosure layer (visibility is defined by contract stakeholders, not by operator privilege).

5.2 Auditability and Selective Disclosure

Confidentiality without accountability is not viable for regulated money movement, and Sotto is designed so the two coexist. Oversight is expressed as an explicit observer relationship: an auditor, regulator, or funder named on a batch receives a complete, real-time, read-only view of the payments it covers — and only those payments. This is the inverse of a mixer or privacy coin, which hides activity from everyone including legitimate oversight; Sotto hides activity from the public while proving it, selectively, to the parties entitled to see it. Identity verification (KYB for payers, KYC for recipients) and sanctions screening are enforced at the authorization layer through integrated providers, so that a mandate cannot disburse to an unverified or screened-out party in the first place.

5.3 Settlement Integrity and Custody

Sotto does not operate a balance sheet and does not take custody of recipients' funds. A payer funds a disbursement in USDCx; recipients receive value into their own Canton accounts, which they alone control. The protocol cannot freeze a recipient's funds or silently reverse a settled payment, because neither operation is reachable from its surface. Atomicity is guaranteed by the network's commit protocol rather than by the operator's reconciliation. For deployments that require operator-side controls — for example, a clawback window under a specific compliance regime — those controls are modeled explicitly as contract terms the recipient can see, never as hidden operator powers.

5.4 Key and Party Management

Each participant in Sotto — payer, recipient, auditor — acts through a Canton party identity authorized by keys held by that participant. Sotto integrates with the network's wallet and custody ecosystem rather than reimplementing key storage, so recipients can hold and control funds through a wallet appropriate to their context, including externally-held keys for institutional payers and lightweight, recovery-friendly accounts for individual recipients in emerging markets.

6. The Integration Layer in Detail

6.1 The Payout Interface

The developer-facing surface of Sotto is a single call that hides the entire lifecycle of a confidential disbursement. The intent shape is intentionally minimal: a reference to the governing mandate, the settlement asset, the disclosure policy, the settlement mode, and the list of payments. Behind it, the protocol resolves recipients, checks each payment against the mandate, constructs the private per-payment contracts, attaches the auditor's observer rights, settles the batch atomically, and returns per-recipient settlement proofs.

```
import { Sotto } from "@sotto/sdk";

const sotto = new Sotto({ orgId: "your-org-id", network: "canton-mainnet" });

const batch = await sotto.disburse({
  mandate: "mandate_2026_q3_payroll",     // on-ledger spending authority
  asset: "USDCx",                          // confidential dollar settlement
  settlement: "atomic",                    // all payments clear, or none do
  observers: ["auditor.acme"],             // permissioned; sees the full batch
  payments: [
    { to: "ada.dev",     amount: "2500.00" },  // private to Ada and the payer
    { to: "kola.design", amount: "1800.00" },  // invisible to Ada, & vice versa
  ],
});

// batch.status      =>  "settled"
// batch.receipts[i] =>  per-recipient settlement proof
```

The developer writes the application; the protocol handles privacy, authorization, and atomic settlement.

6.2 Atomic Batch Settlement

A payout is rarely a single payment. Sotto treats a batch as a first-class unit: the payer chooses whether the batch settles atomically — every payment clears or the whole batch rolls back — or independently, where each payment settles on its own. Atomic mode removes the reconciliation burden of partial failure entirely, which is decisive for payroll and scheduled disbursements where “some people got paid and some didn't” is an operational emergency.

6.3 Off-Ramp and Last-Mile Settlement

USDCx is the rail's settlement asset; it is not necessarily the form in which a recipient wants to hold value. Sotto connects to local off-ramp partners at the receiving end, so a recipient in an emerging market can convert a confidential dollar payment into local currency in their bank account or mobile-money wallet, at fees set by the off-ramp rather than by the protocol. Sotto owns the confidential payout-and-audit layer and composes with the existing universe of regional cash-out providers rather than rebuilding it.

6.4 Data Import and Canton-Native Surfaces

To meet organizations where they already work, the integration layer ingests payout instructions from the systems that produce them — payroll and HR exports, marketplace ledgers, grant-management spreadsheets — and translates them into mandate-governed batches. Applications interact with the rail through Canton's JSON ledger API, so a back office, a web dashboard, or a mobile recipient app can all invoke the same protocol against the same accounts.

7. Surfaces of the Protocol

Sotto is not a single application. It is a layer that several surfaces invoke, none privileged over the others.

The reference application is the first surface to ship: a working payout console, built by Sotto, that exercises the entire protocol end to end. A payer creates a mandate, imports or enters a batch, runs an atomic confidential disbursement, and watches it settle; a recipient sees only their own payment; a named auditor sees the whole batch. It is a fully functional product, and also the proof-of-protocol artifact everyone who builds on the rail can evaluate.

The developer SDK lets any third-party application invoke the protocol from inside its own product — a marketplace embedding payouts, a fund automating distributions, an HR platform adding confidential salary settlement. The SDK ships for TypeScript and Python on the back end, with a Flutter client for recipient-facing mobile experiences, each exposing the same interface against the same accounts.

Partner integrations connect the rail to the ecosystem at both ends: identity and sanctions providers at the authorization layer, and regional off-ramp partners at the settlement layer, so that an organization can adopt Sotto without separately assembling a compliance stack and a cash-out network.

8. Why Sotto Is Different

Versus transparent-chain payout rails. The current generation of stablecoin payout infrastructure is fast and inexpensive, and it treats the transparency of the underlying chain as a feature — an on-chain record as a single source of truth. For an organization paying people at scale, that same transparency is the liability: the payout ledger is public. Sotto delivers the same speed and cost on a ledger that is confidential by default and provable by permission. Privacy is not something a developer bolts on; it is what the rail is.

Versus traditional payout providers. Conventional payroll and disbursement platforms keep payout data private, but inside a custodial silo, settled over slow correspondent rails, in batches that fail piecemeal, with privacy that an auditor must take on trust. Sotto does not operate a balance sheet, settles atomically in seconds, and replaces opaque privacy with confidentiality that can be selectively and cryptographically proven to exactly the right parties.

Versus mixers and privacy coins. Approaches that achieve on-chain privacy by hiding activity from everyone — including legitimate oversight — are incompatible with regulated money movement and tend to break composability. Sotto is the opposite construction: activity is confidential to the public and the competition, and transparent, in real time and read-only, to a named auditor or funder. It is privacy *with* accountability, which is the only kind that scales into finance.

The cumulative effect is that Sotto occupies a layer the existing categories do not. It is not a better transparent rail, a faster bank, or a more private mixer. It is the confidential payout layer that, once present, lets every kind of organization disburse at scale without publishing what it is doing.

9. Use Cases

Confidential payroll. A company pays its team in USDCx; each employee sees only their own pay, the company's compensation structure never touches a public ledger, and a payroll auditor sees the complete run.

Marketplace and platform payouts. A marketplace pays its sellers or contractors without revealing seller earnings, the platform's take rate, or its roster to competitors — while retaining a provable internal record.

Cross-border contractor payments. An organization pays remote contractors in emerging markets confidentially in dollars, settling in seconds, with a local off-ramp converting to the recipient's currency at the receiving end — the lead wedge for the protocol, and the corridor where speed, cost, and privacy compound most.

Vendor and supplier disbursement. A business pays suppliers without exposing per-vendor terms to other vendors or to the market, with atomic settlement removing partial-payment risk.

Aid and grant disbursement. An NGO or foundation disburses to beneficiaries whose identities must be protected, while granting its donors and regulators a real-time, read-only view that proves funds reached their intended recipients — a property no transparent chain can offer, because recipient privacy and public auditability are mutually exclusive on a transparent ledger.

Fund and treasury distributions. A fund distributes to its investors or portfolio companies confidentially, with positions and allocations kept private between the fund and each counterparty.

10. Roadmap

The protocol ships in four phases. Each phase exposes a new surface above the same protocol core and broadens the set of organizations that can invoke it. The sequencing follows the go-to-market logic: prove the rail, win a commercial beachhead, open it to developers, then extend into the mission verticals where the privacy guarantee matters most.

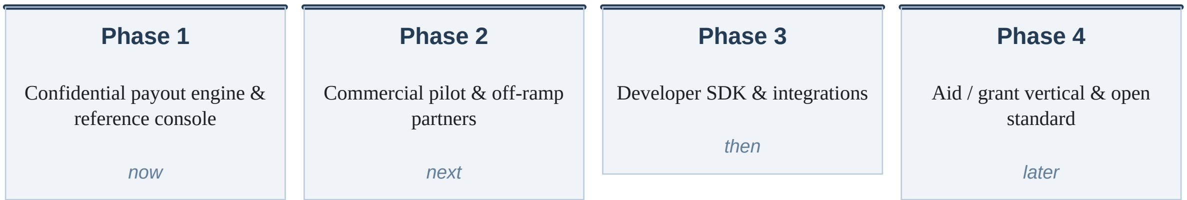

Figure 2. The four-phase roadmap. Each phase is additive; nothing shipped earlier is replaced by what comes later.

Phase 1 — Confidential Payout Engine and Reference Console (now). The protocol core ships on Canton: mandate-governed authorization, private per-recipient payments, the permissioned auditor view, and atomic batch settlement, demonstrated through a reference console that runs the full flow. This is the proof-of-protocol artifact, and the foundation everything else is built on.

Phase 2 — Commercial Pilot and Off-Ramp Partners. Sotto onboards a design-partner payer — a marketplace, agency, or cross-border employer — integrates a regional cash-out partner for local settlement, and moves disbursement onto USDCx on the live network, with KYB, KYC, and sanctions screening enforced at the authorization layer.

Phase 3 — Developer SDK and Integrations. The protocol becomes consumable by any application through the SDK and the payout interface, with importers for payroll, HR, and marketplace data, and additional cash-out corridors. Organizations integrate confidential payouts without rebuilding the underlying machinery.

Phase 4 — Aid and Grant Vertical, and an Open Standard. Sotto extends into confidential disbursement for NGOs and funders with the donor-observer pattern, broadens to institutional deployments, and works toward contributing its core Daml templates as an open, conformance-tested standard so the confidential payout primitive can outlast any single operator.

11. Conclusion

The difficulty of paying many people across borders, privately, at scale, is not a blockchain problem. Stablecoins already settle in seconds at near-zero cost, and Canton already keeps transactions private from non-parties, settles them atomically, and admits a permissioned auditor. What is missing is the rail that turns those capabilities into a single, callable primitive for confidential disbursement.

Sotto builds that rail. By making confidentiality a property of the settlement layer rather than a feature an application struggles to add on top — and by pairing it with on-ledger authorization, atomic settlement, and selective, provable auditability — the protocol lets a company, a marketplace, a fund, or an aid agency pay everyone it needs to pay without publishing whom it pays, how much, or how often, and while still proving every payment to the parties entitled to see it.

The applications are where the protocol becomes visible. The rail is where the confidentiality lives.

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